

ACE REAL ESTATE OS — V1 ARCHITECTURE DOCUMENT

1. SYSTEM OVERVIEW

ACE is a multi-tenant real estate lead orchestration system combining:

- Lead intelligence engine
- Property inventory system
- WhatsApp conversation + nurture engine
- Broker workflow system
- Billing + wallet system

2. CORE STATE MACHINES

LEAD STATES:

NEW, IN_CONVERSATION, QUALIFYING, QUALIFIED,
PROPERTY_SHOWN, INTERESTED, VISIT_REQUESTED,
AWAITING_BROKER_APPROVAL, VISIT_CONFIRMED, VISIT_COMPLETED,
INACTIVE_24H, INACTIVE_3D, INACTIVE_7D, DORMANT,
RESURRECTED, LOST, CONVERTED

BROKER STATES:

ONLINE, OFFLINE, HUMAN_MODE_ACTIVE, UNRESPONSIVE, SLA_BREACHED

PROPERTY STATES:

ACTIVE, RESERVED, SOLD, RENTED, OFF_MARKET

SESSION MODES:

AI_ACTIVE, HUMAN_ACTIVE

3. HUMAN MODE RULES

- Broker TAKEOVER disables bot completely
- Auto resume = last broker message + 1 hour
- Resume notification sent via email to broker/admin only

4. NURTURE ENGINE

High Intent:

3h, 6h, 12h, 21h (before 10 PM cutoff)

Short Term:

36h, 3 days, 7 days

Long Term:

Random 2–5 day intervals

Decay:

1 month weekly

2 months bi-weekly

3+ months archived low priority

Meta Rule:

Any user reply resets 24-hour free WhatsApp window

5. WALLET SYSTEM

- Prepaid ledger-based system
- Per message debit
- Cost + markup tracking

States:

HEALTHY, LOW, CRITICAL, BLOCKED

Alerts ONLY to brokers/admin, NOT customers.

6. BROKER SLA SYSTEM

15 min internal alert

2 hr escalation

6 hr SLA breach

24 hr unresponsive

7. PROPERTY LIFECYCLE

ACTIVE, RESERVED, SOLD, RENTED, OFF_MARKET

Manual updates in V1

8. RELIABILITY ENGINE

- Idempotency keys prevent duplicate processing
- Webhook deduplication
- Queue-based retry system
- Event persistence mandatory
- Degraded mode on outages

9. TAKEOVER SYSTEM

- Only one controller: AI or HUMAN
- Human mode freezes bot actions
- Auto timeout after inactivity + 1 hour

10. SUPERADMIN DASHBOARD

- System health (MSG91, Meta)
- Client health (wallet, SLA)
- Revenue metrics (MRR, ARR, ROI)
- Funnel analytics
- Risk monitoring

11. CORE PRINCIPLES

- Event-driven architecture
- State machine driven logic
- Human-in-loop for approvals
- Cost-aware messaging system
- Zero data loss guarantee